

Leo Wang

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Skills

- Languages ◇ **Advanced:** C#, Python
Intermediate: TypeScript/JavaScript, HTML/CSS, Java, C++
Beginner: HLSL, Matlab
- Technologies ◇ **Platforms:** Unity, NextJS
Libraries: Dear ImGui, Newtonsoft, OpenCV, Numpy, Selenium, Tensorflow, Matplotlib, TailwindCSS, React, Vite
Dev. Tools: Visual Studio, Android Studio, Visual Studio Code, Git, Github
- Skills ◇ Software Optimization, Large-Scale Data Processing, Data Visualization, Web Development

Education

- 2024-26 ◇ **University of Maryland, Bachelor of Science.**
 - Double Major: **Computer Science** (Data Science) and **Math.**
 - GPA: 4.0
 - Advanced Cybersecurity Experience for Students (ACES) Honors College. (Previous)
 - Computer Science Departmental Honors Program. (Current)
- 2024 ◇ **UMD Presidential Scholarship** (\$5,000/year)
- 2024 ◇ **National Merit Scholarship** (\$1,000/year)

Experiences

- 2024 ◇ **Cybersecurity Training Platform for MITRE Corporation.** Web Developer, UMD
 - I worked with a student led team to develop a user-friendly yet powerful cybersecurity training platform for new MITRE employees. I built the frontend section of the admin-side pages using NextJS, TailwindCSS and Typescript.
- 2023 ◇ **IEEE Aerospace Conference.** Co-Author, Virtual
 - "Enhancing Space Communications: A Novel Approach to Solving the Multi-Satellite Scheduling Problem"
 - Published paper on work done during the NASA internship to the 2024 IEEE Aerospace Conference.
- 2023 ◇ **IEEE Integrated STEM Education Conference.** Presenter, Johns Hopkins Applied Physics Lab
 - "Enhancing STEM Education to Communities with Low Access to STEM"
 - Presented at IEEE Integrated STEM Education Conference on raising STEM engagement in under-served communities.
- 2021-23 ◇ **National Aeronautics and Space Administration (NASA).** Intern, Goddard Space Flight Center
 - I led a team of 5 interns as lead developer to create a high fidelity and performant planetary terrain/orbiting satellite simulation that outperformed existing tooling in C# using the Unity Engine. (See Projects)

Projects

- 2024 ◇ **Personal Website.**
 - Designed and implemented a fully custom personal website using React, TailwindCSS, and Vite.
- 2024 ◇ **N-Body (Barnes-Hut) Simulation.**
 - Developed an optimized N-body simulation (**n=15,000**) by leveraging the Barnes-Hut algorithm in C++ and Dear ImGui.
- 2022-23 ◇ **NASA Terrain System.**
 - I created an extremely powerful planetary terrain system that was the backbone of my NASA internship, capable of dynamically loading terrain data at arbitrary resolutions in realtime.
 - The internal database stores highly compressed height maps ranging from **tens of millions to billions of points**. (ex. one region stores 170 million points in 20 mib).
 - The system can process, render, and simulate up to **13,000,000 points per second** (including database read times).